Inspire the love of learning

HW1 - 2353235 - Phạm Minh Trí

A1. Frame a learning problem

- Input: scan of the address block of the envelope. (x)
- Target: the destination sorting bin / regional sorting center, one of K bins, determined by the postal code (y)
- Decision: send envelope to that bin, if low confidence -> human operator
- Metric: top-1 routing accuracy (misroute rate)
- Label noise: operator mis read bad handwriting => wrong bin label
- One reason the same data supports classification or regression:
  + If the decision is which bin to send the envelop to, then y is a discrete label, so this is classification (measured by accuracy)
  + If the decision is when to promise delivery, then y is a real number like hours until deliver -> regression (MAE / RMSE)

A2. Evaluation under limited data

(i): 80/20: 96 train, 24 test.

- What it tells: an estimate of how a model trained on 96 examples performs on 24 unseen example
- Does not tells: how stable this number is. With 24 examples, 1 extra error -> 1/24 = 4.2%, and a different random split could give a very different score. No tuning without touching the test set.

(ii): 5-fold CV on training portion + final untouched test set.

- What it tells: the average validation score & how much it varies across folds, so we can compare models & choose hyperparameters. The untouched test set give one honest final estimate of the chosen model.
- Does not tells: test score is unbiased but still noisy because it has only 24 examples. Neither they says anything about performance if the deployment data shift (new handwriting / seasons)

• Why use test set repeatly: If we try many hyperparameter

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settings & keep the one with the best test score, we pick the maximum of several noisy scores, and the maximum of noisy score is on average higher than the true performance (selection bias). The test set then influenced the choice of model → no longer independent data. The fix is to tune on validation / CV & use the test set only once at the end.

A3. Metrics & decision context.

- Given: TP = 36, FN = 14, FP = 30, TN = 170. Total N = 250

- Accuracy = $\frac{TP+TN}{N} = \frac{36+170}{250} = 0.824$

- Precision = $\frac{TP}{TP+FP} = \frac{36}{36+30} = 0.5455$

- Recall = $\frac{TP}{TP+FN} = \frac{36}{36+14} = 0.72$

- $F_1 = \frac{2 \cdot P \cdot R}{P+R} = \frac{2TP}{2TP+FP+FN} = \frac{72}{72+30+14} = 0.6207$

- Why accuracy can mislead: when positive class is rare, a model that always predict negative still get a high accuracy here is 200/250 = 0.8, so the classifier's 0.824 is barely better than doing nothing. (Positive = TP + FN = 50)

- Error type needing special attention: the false negative (real event missed) because the event is rare but costly, while a false positive only cause an extra check. So we should watch recall (14 of 50 are missed) and choose the decision threshold using the cost of FN vs. FP.

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A4. Diagnose model behaviour

-Model A (24%, 26%). Diagnose: underfitting (high bias), both error high, gap only 2%. Intervention: increase capacity, add features, reduce regularization.

-Model B (3%, 21%). Diagnose: overfitting (high val), train very low, gap 18%. Intervention: regularise, simplify the model / get more training data.

-Model C (9%, 11%). Diagnose: healthy, both errors low, gap only 2%. Intervention: keep.

• Complexity) bias and variance: When complexity increase, the model can fit more complicated pattern so bias ↓ but it also start fitting noise in particular training set =) variance ↑. Training error keeps falling while validation error first fall and then rise. The best model is near the lowest point of validation curve.

A5. Assignment bridge: protocol card.

-Frozen target: the activity label with 6 classes (walking, walking upstairs, walking downstairs, sitting, standing, laying). The definition is fixed & will not change after seeing results.

-Split policy: subject-disjoint. Official split: 21 subjects for training (7352 windows), 9 subjects for test (2947 windows). Model selection uses GroupKFold(5) grouped by subject ID on training part, seed 42. The test set is used once at the end.

-Primary metric: macro-F1 over 6 classes. It gives equal weight for each class, so mistakes between similar static posture (sitting vs. standing) are not hidden.

-One leakage risk: windows overlap by 50% & neighbouring windows come from the same person. A random window-level split puts near duplicate windows in both train & test, inflate the score. Scalers must also be fitted on training folds only.

-Comparability rule: all models use same folds, preprocessing, seed, metric, tuning budget.

-Limitation: only 30 volunteers, 1 phone type worn at the waist. New user/placement need new data.

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